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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the essential elements of an algorithm?

Answer (Please click here to Add Answer)

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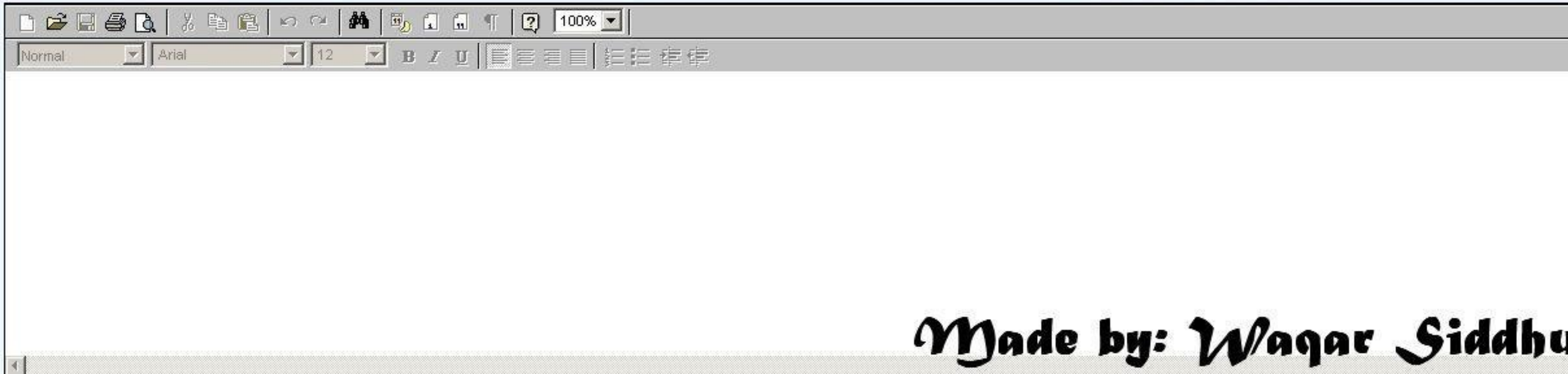

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What is overall time for Kruskal's algorithm if the graph is sparse?

Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is DFS and how does it work?

Answer (Please click here to Add Answer)

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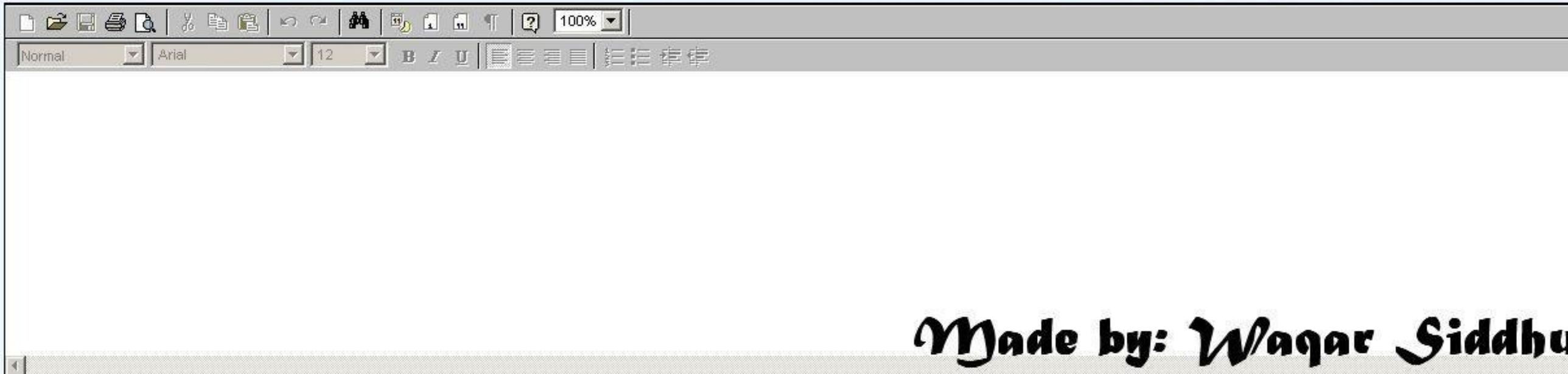

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

What is Bellman-Ford algorithm's running time?

Answer (Please click here to Add Answer)

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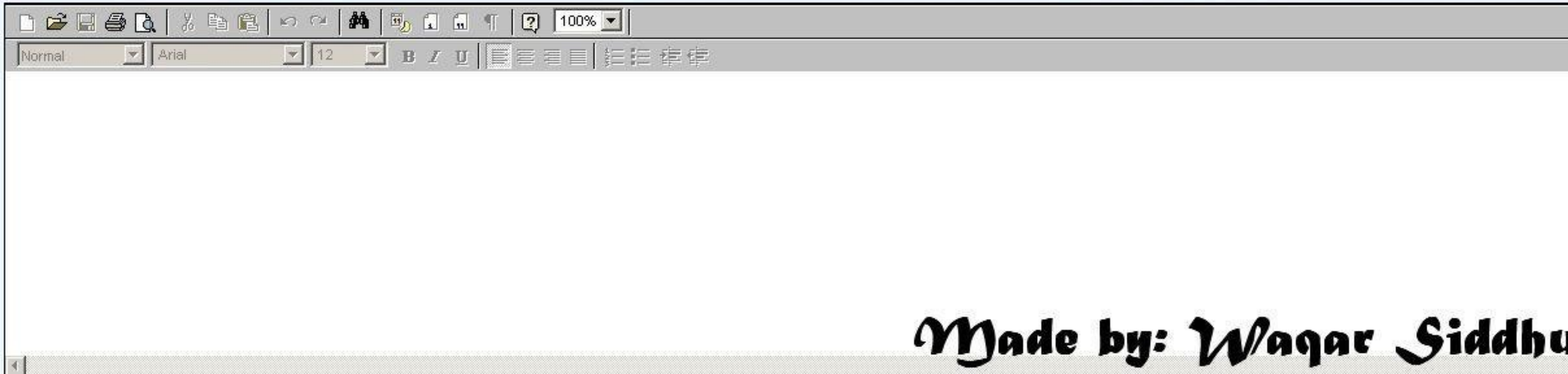

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Explain the following two basic cases according to Floyd-Warshall Algorithm,
1. Don't go through vertex k at all.
2. Do go through vertex k.

Answer (Please click here to Add Answer)

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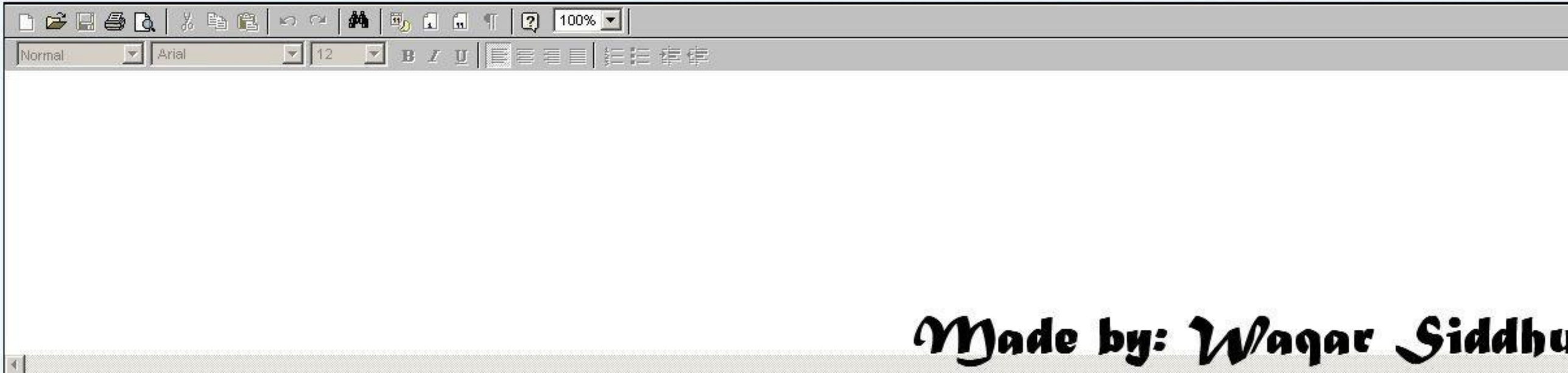

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Describe 2-d maxima problem in mathematical or algorithmic form.

Answer (Please click here to Add Answer)

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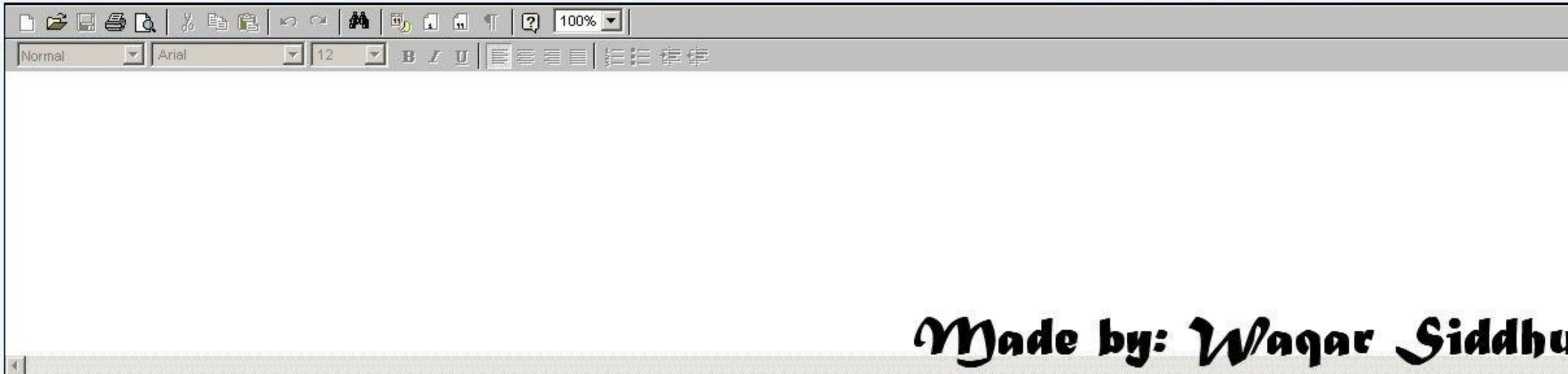

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Describe Minimum Spanning Trees Problem with examples.

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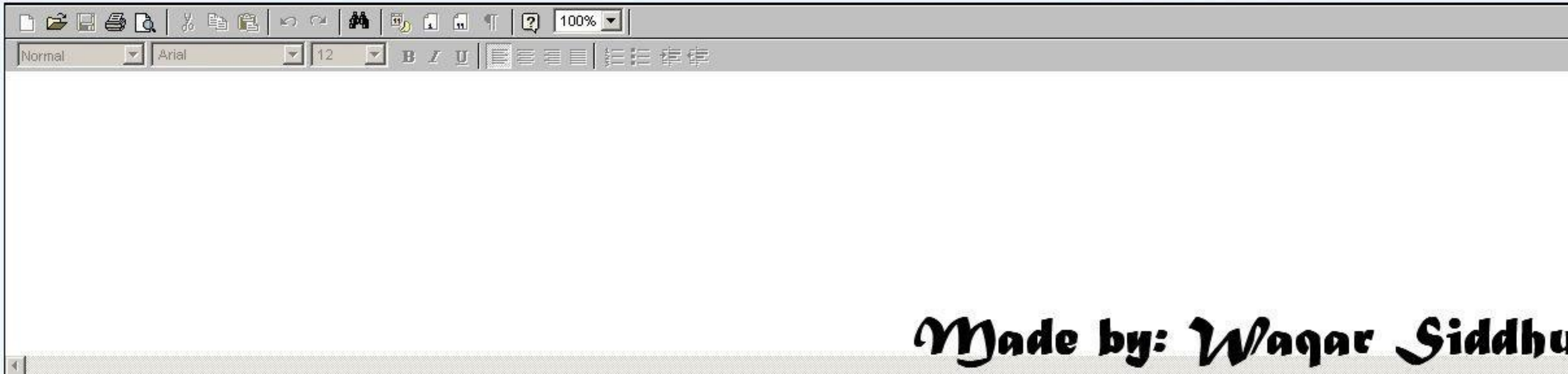
Answer (Please click here to Add Answer)



Let the adjacency list representation of an undirected graph is given below:
Is there any isolated vertex? If yes, Name the vertex.
What general property of the list indicates that the graph has an isolated vertex?

a → b → c → e
b → a → d

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Answer (Please click here to Add Answer)

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a → b → c → e
b → a → d
c → a → d → e → f
d → b → c → f
e → a → c → f

Answer (Please click here to Add Answer)

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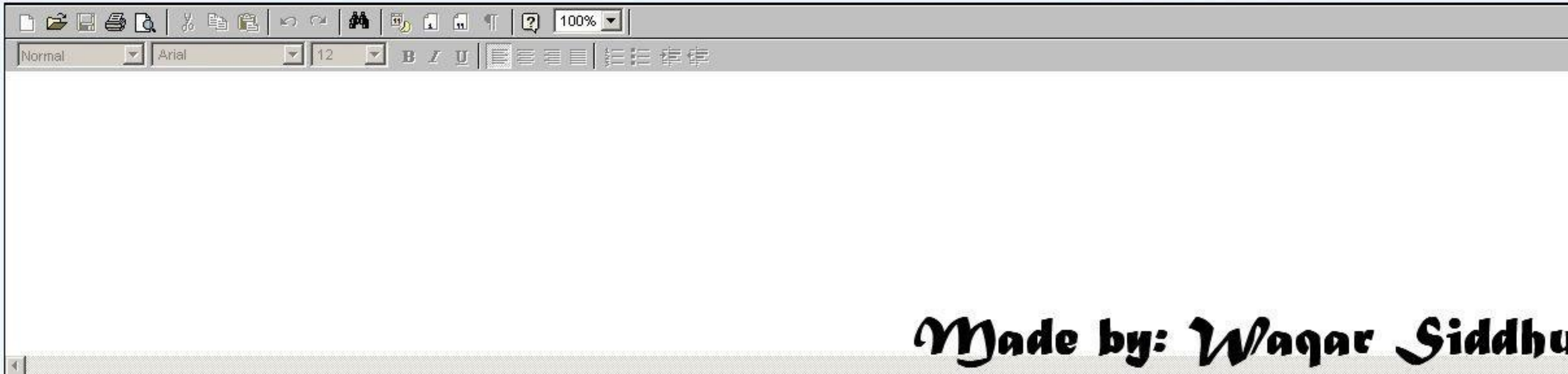

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b → a → d
c → a → d → e → f
d → b → c → f
e → a → c → f
f → c → d → e
g

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Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

c → a → d → e → f
d → b → c → f
e → a → c → f
f → c → d → e
g

Answer (Please click here to Add Answer)

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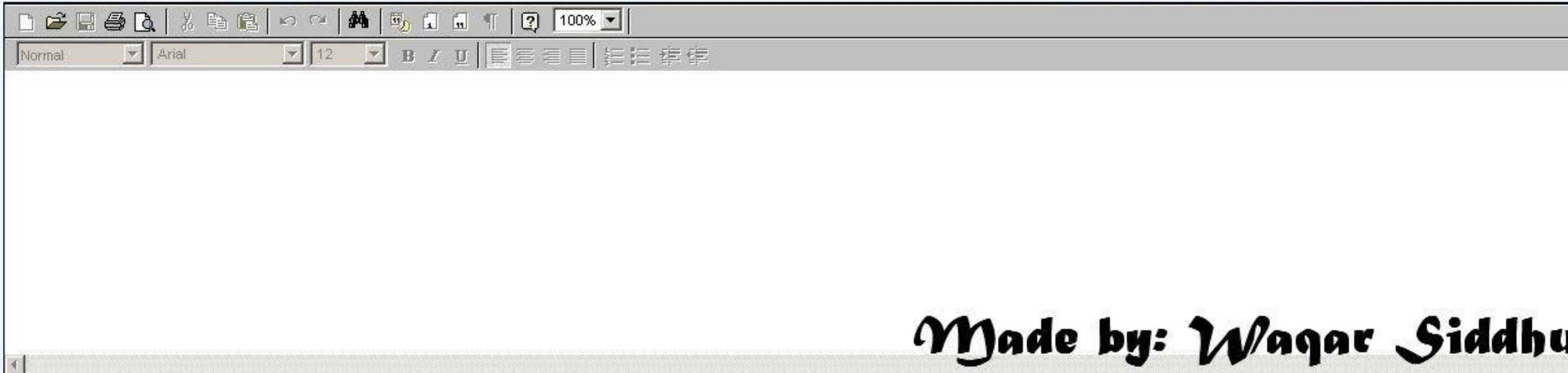

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You are given the task of laying down new railway lines which will connect all *n* cities. Thus for any pair of cities, you will end up with track connecting them. Note that two routes may share the same track; track laid between Lahore and Islamabad can be used to travel in both directions. Your goal is to use the minimum amount of track. How would you achieve the goal now? (Note : consider the scenario carefully and name only the best suited algorithm)

1 Dijkstra's algorithm
2 Prims Algorithm
3 Folloyed Warshal Agorithm

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Answer (Please click here to Add Answer)

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track; track laid between Lahore and Islamabad can be used to travel in both directions. Your goal is to use the minimum amount of track. How would you achieve the goal now? (Note : consider the scenario carefully and name only the best suited algorithm)

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2 Prims Algorithm
3 Folloyed Warshal Agorithm
4 Bellman Ford Algorithm.

Answer (Please click here to Add Answer)

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Show the result of time stamped DFS algorithm on the following graph. Take node E as a starting node. [You can show final result in exam software and need not to show all intermediate steps].

Answer (Please click here to Add Answer)

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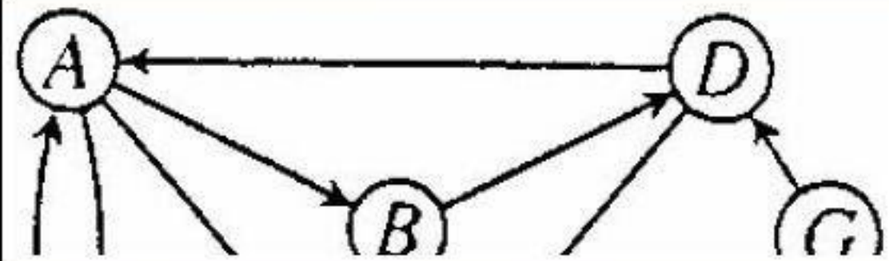

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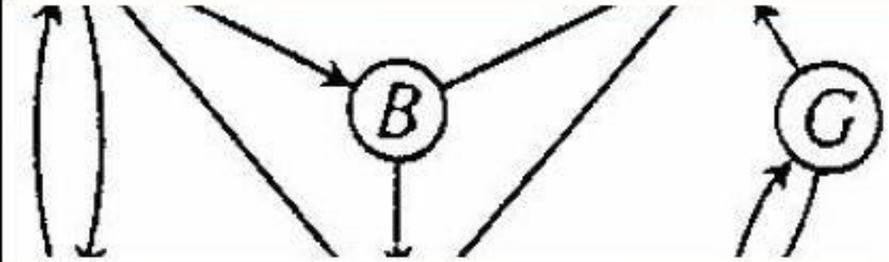

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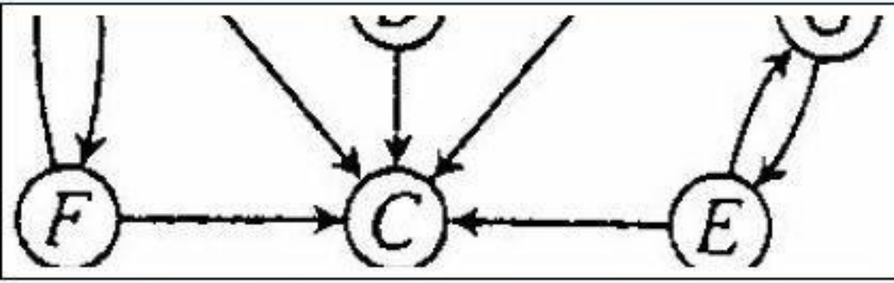

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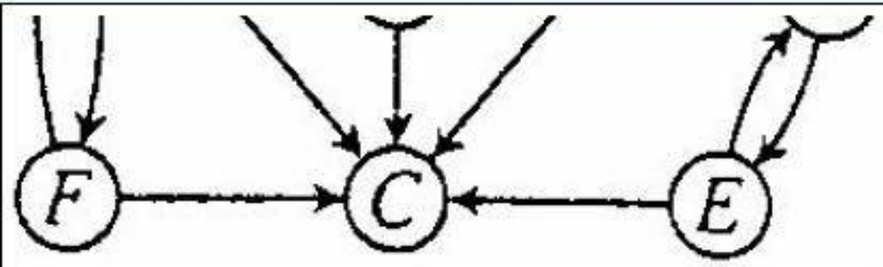

Answer (Please click here to Add Answer)

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Made by: Waqar Siddhu

Develop the running time complexity analysis for the following piece of code. Adopt step wise approach along with asymptotic notation at the end.

```
i=1
while (i < n) {
   i++
}
for ( i=1;i <= n ;i=i*2 )
```

VuAnswers.com

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

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```
i=1
while (i < n) {
   i++
}
for ( i=1;i <= n ;i=i*2 )
for ( j = 1; j <= i; ++j )
```

Answer (Please click here to Add Answer)

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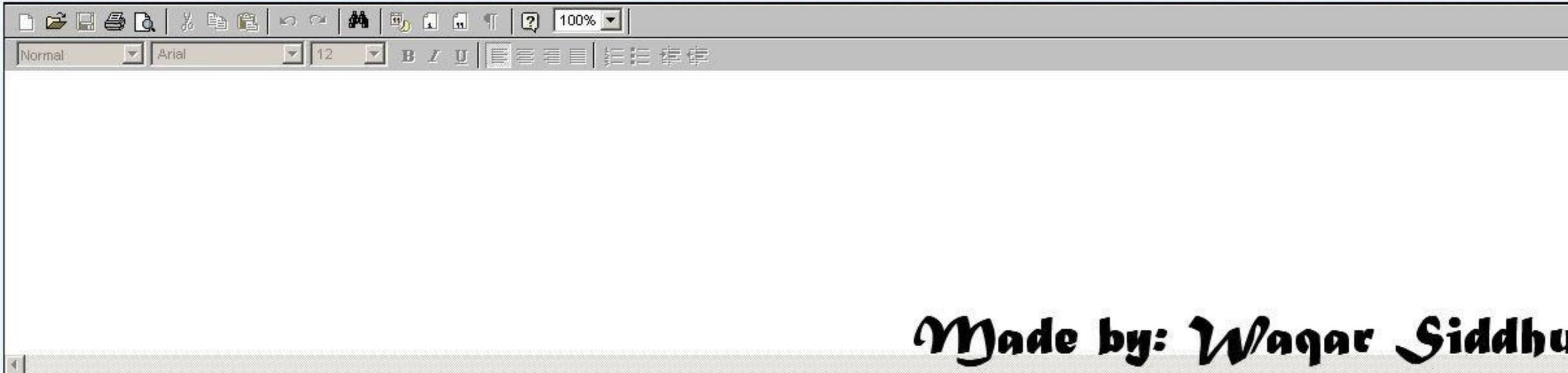

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Kruskal's algorithm can return different spanning trees for the same input graph G depending upon how ties are broken when edges are sorted. Prove that, for each minimum spanning tree T of G, there is a way to sort the edges of G in Kruskal's algorithm so that the algorithm returns T.

Answer (Please click here to Add Answer)

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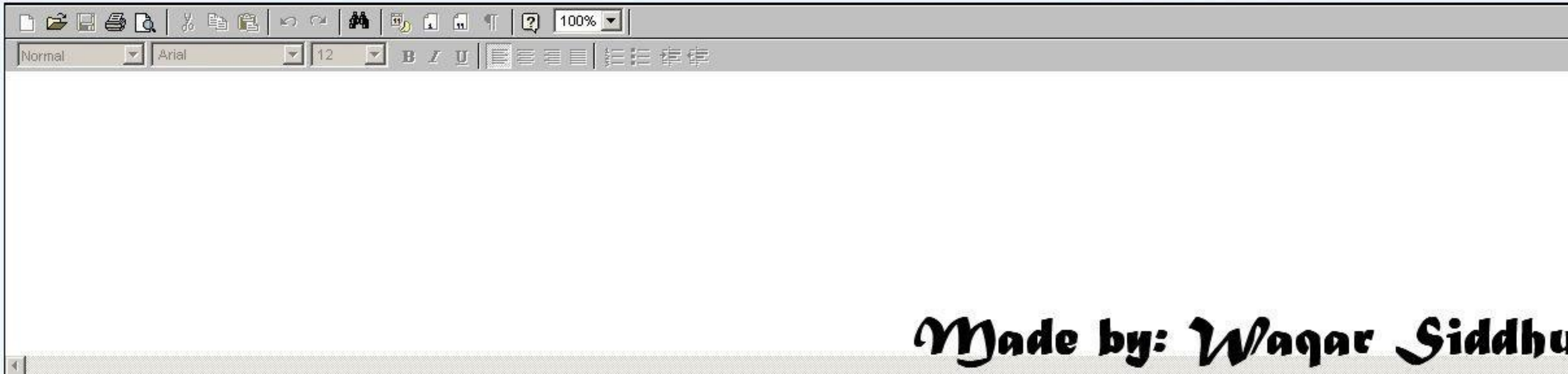

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

How we Heapify?

Answer (Please click here to Add Answer)

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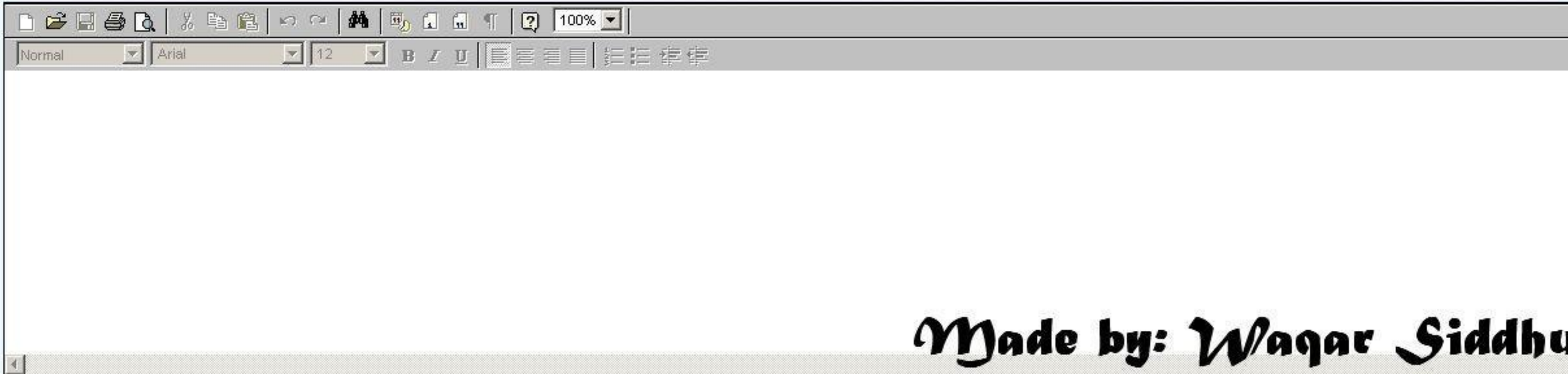

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Define Back Edge

Answer (Please click here to Add Answer)

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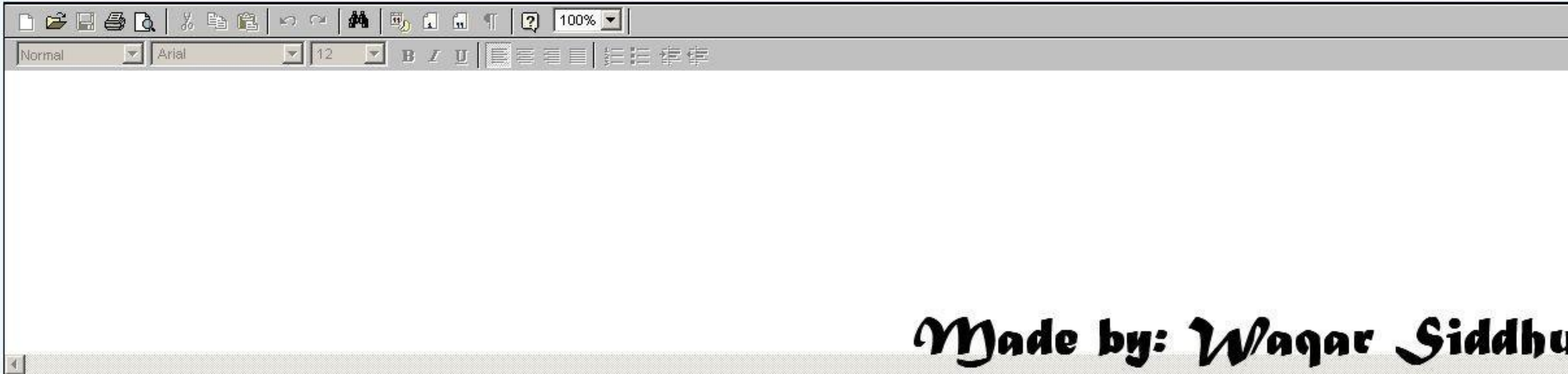

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Given an adjacency list for G, what is the time complexity to compute G^T.?

Answer (Please click here to Add Answer)

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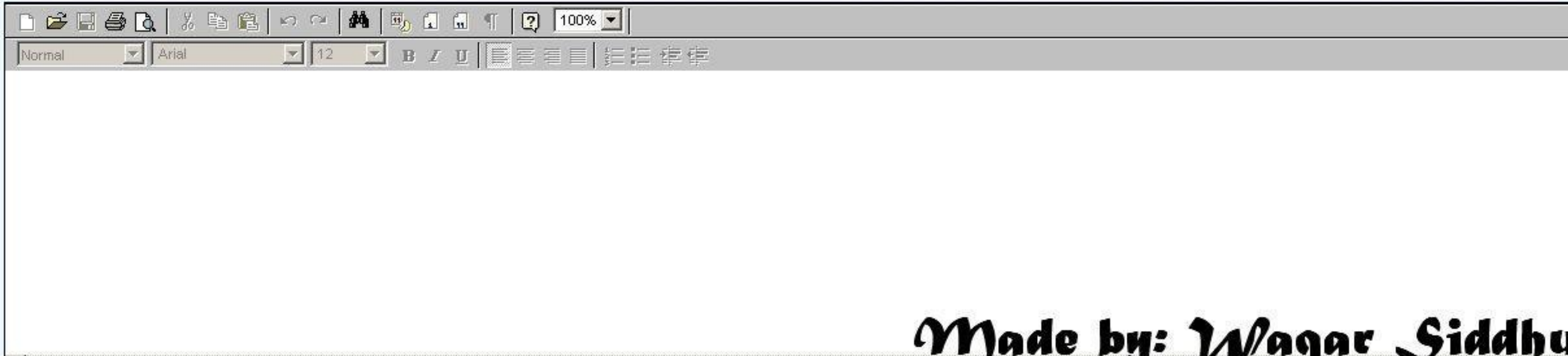

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

What is Bellman-Ford algorithm's running time?

Answer (Please click here to Add Answer)

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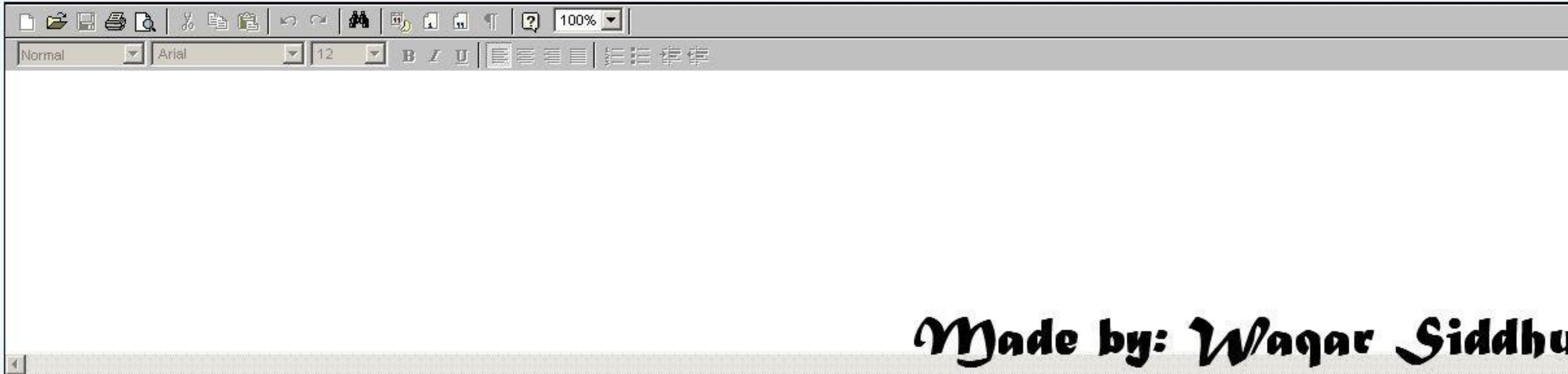

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Given a digraph G = (V, E), consider any DFS forest of G and consider any edge (u, v) ∈ E. Prove that If this edge is a tree, forward or cross edge, then f [u] > f [v] and if this edge is a back edge, then f [u] ≤ f [v].

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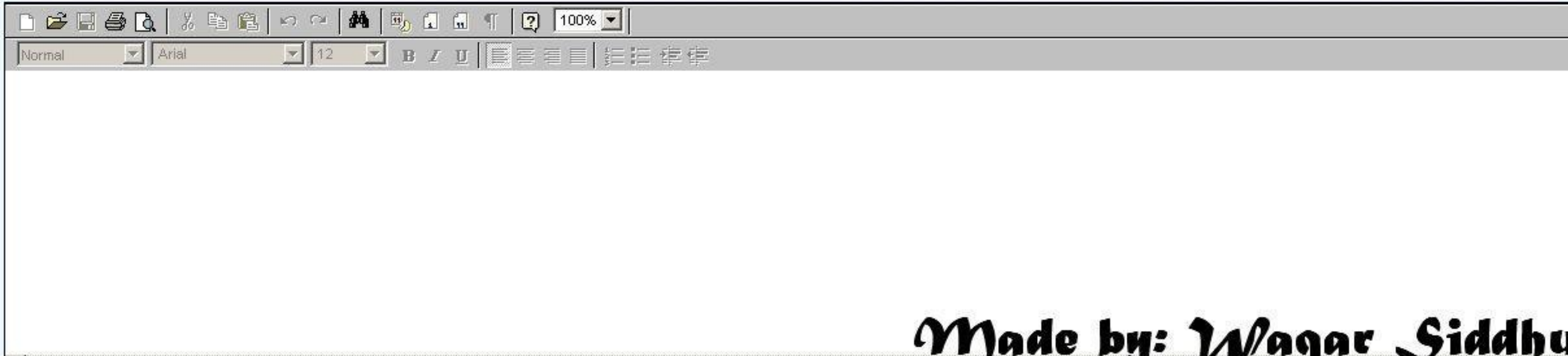
Answer (Please click here to Add Answer)



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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

How the Dijkstra's algorithm works?

Answer (Please click here to Add Answer)

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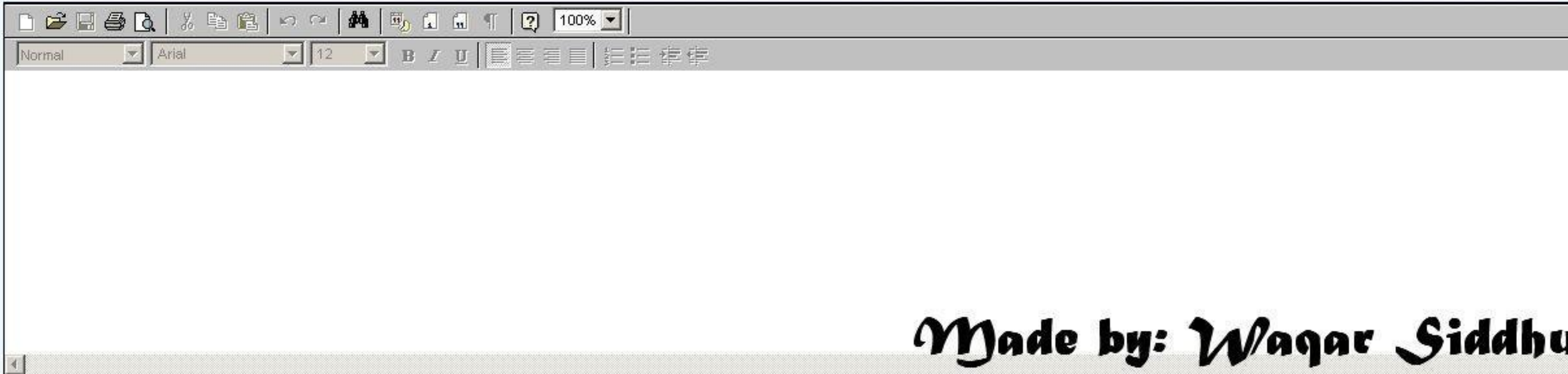

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Modify QUICKSORT algorithm such that it sorts array into non-increasing order.

Answer (Please click here to Add Answer)

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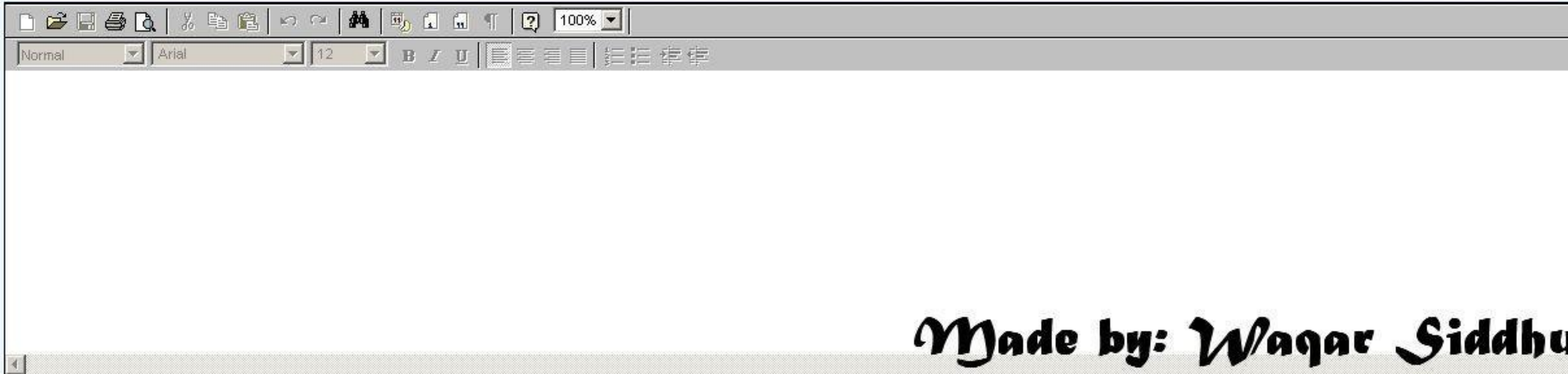

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What do you mean by polynomial time algorithm? Explain what kind of problem can be solved by using polynomial time algorithm?

Answer (Please click here to Add Answer)

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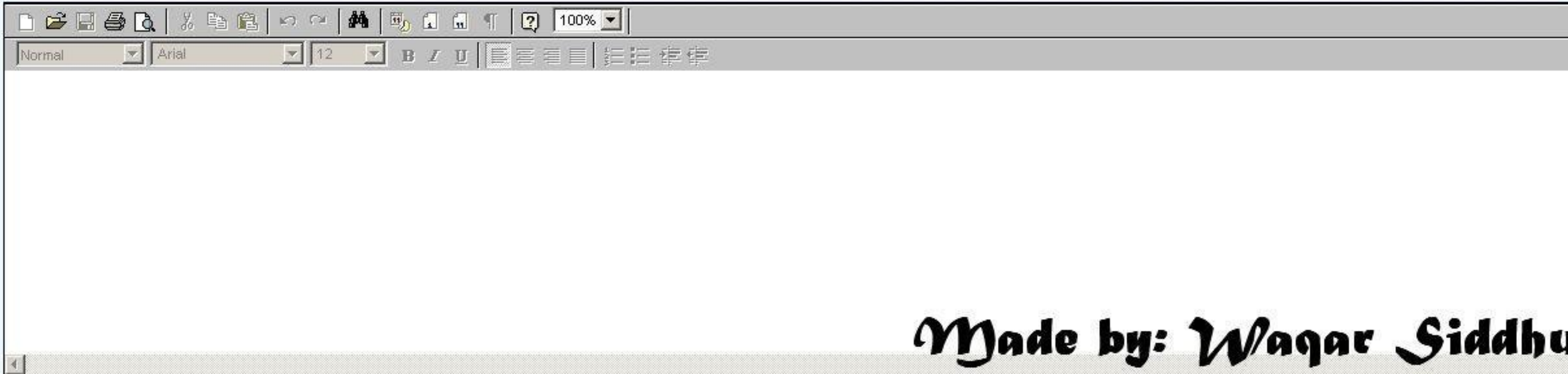

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2 Prims Algorithm
3 Folloyed Warshal Agorithm

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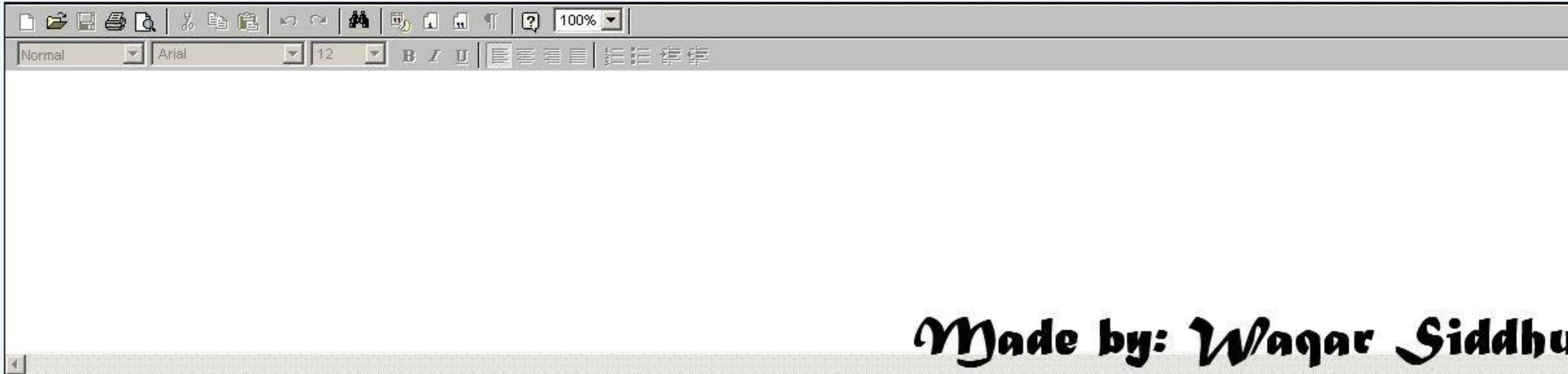
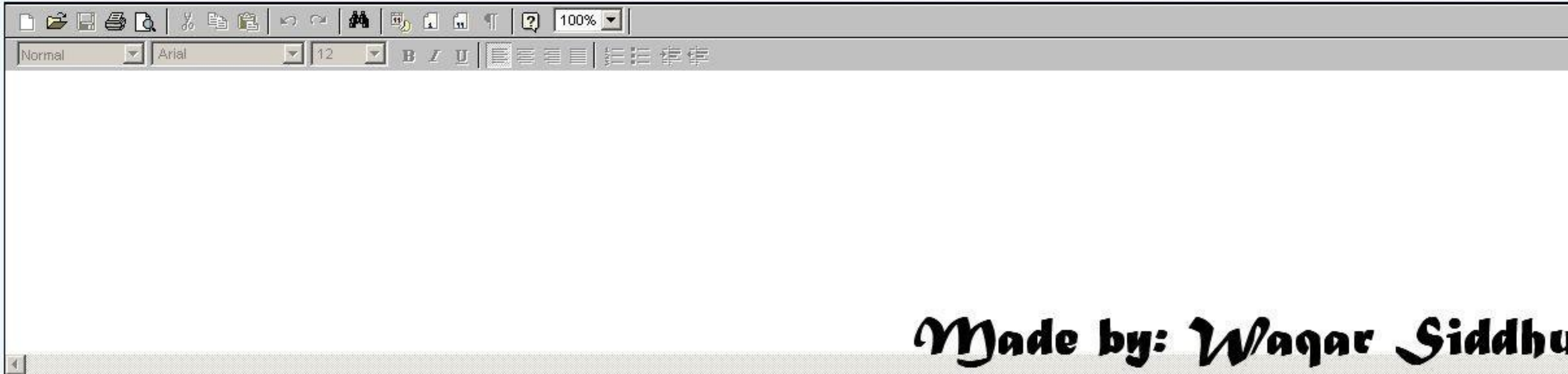


Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Considering the recursive version of depth-first traversal implementing Timestamp Structure in pseudo code format, only write DFSVISIT routine in pseudo code format

Answer (Please click here to Add Answer)

VuAnswers.com



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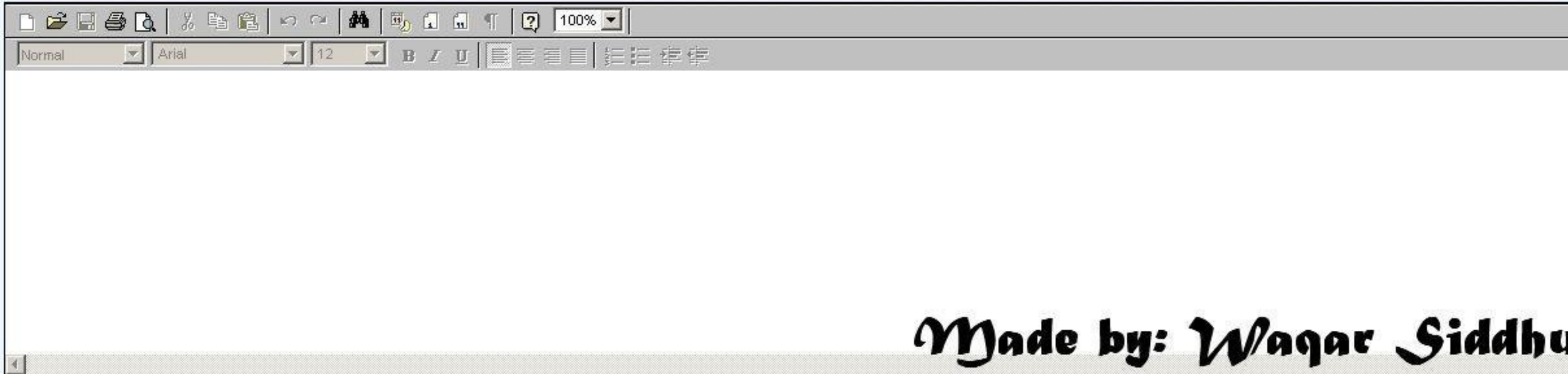

Made by: Waqar Siddhu

Develop the running time complexity analysis for the following piece of code. Adopt step wise approach along with asymptotic notation at the end.

```
i=1
while (i < n) {
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}
for ( i=1;i <= n ;i=i*2 )
```

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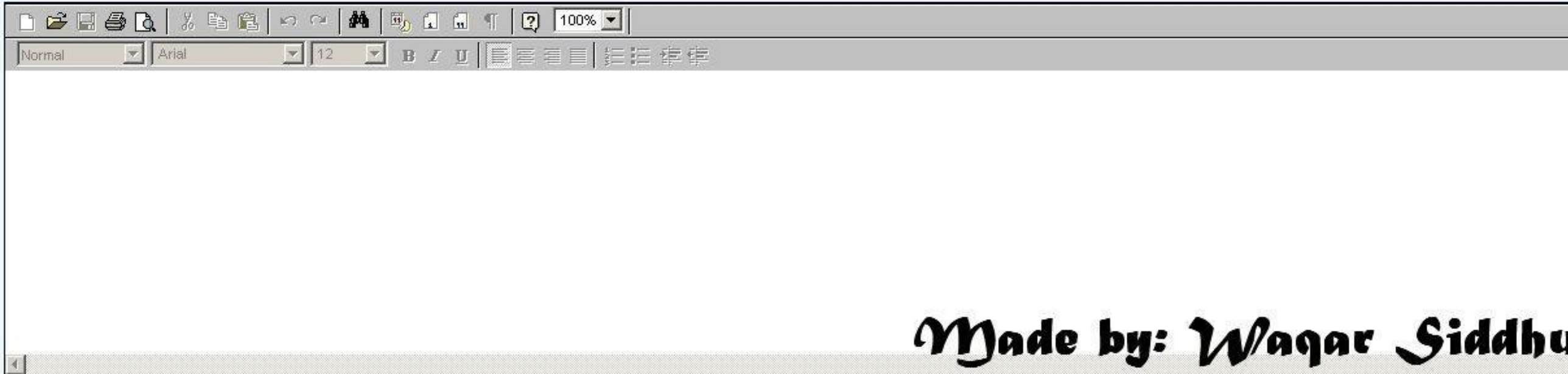
VuAnswers.com



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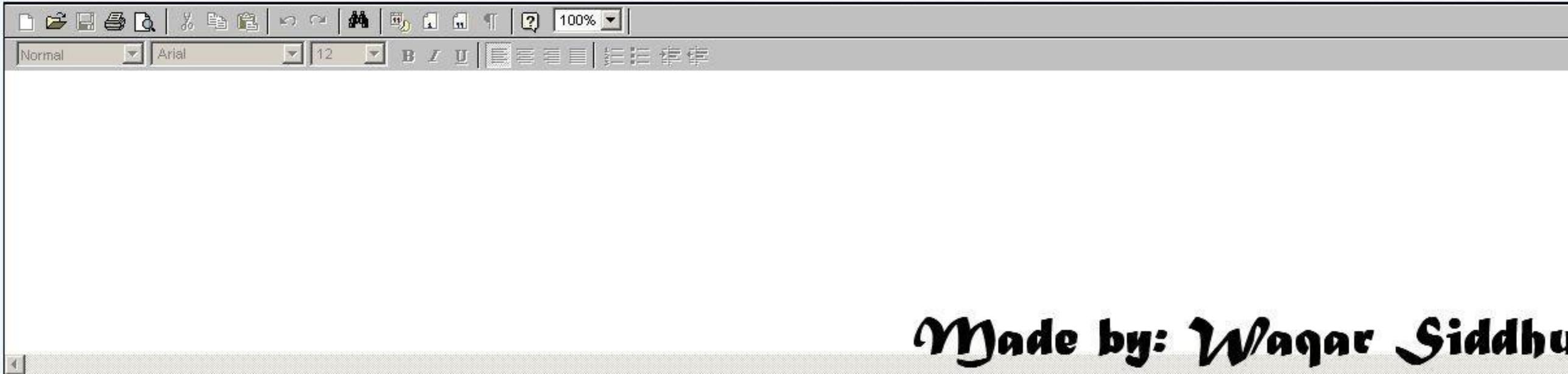

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Write pseudo code for Kruskal's algorithm.

Answer (Please click here to Add Answer)

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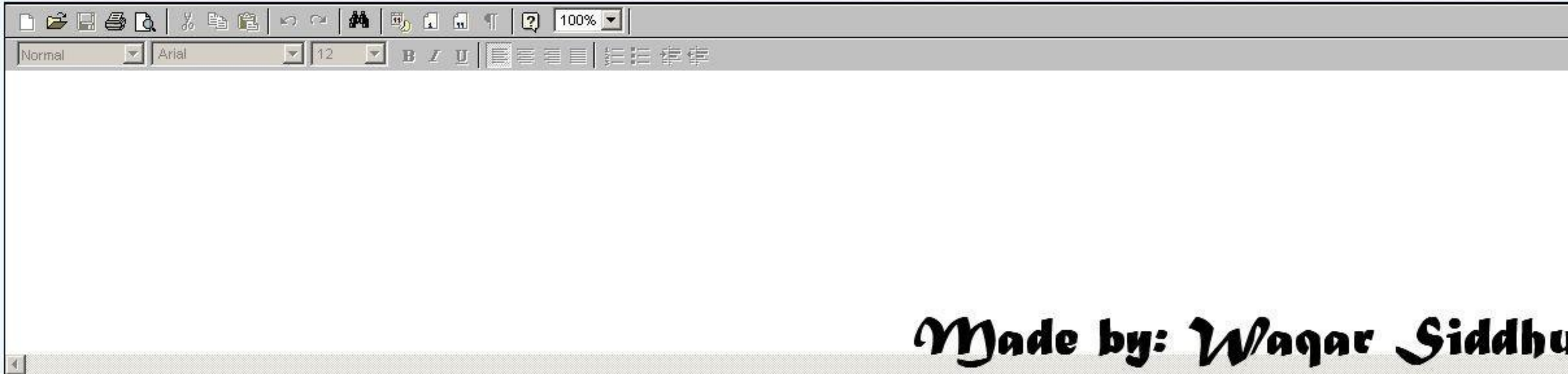

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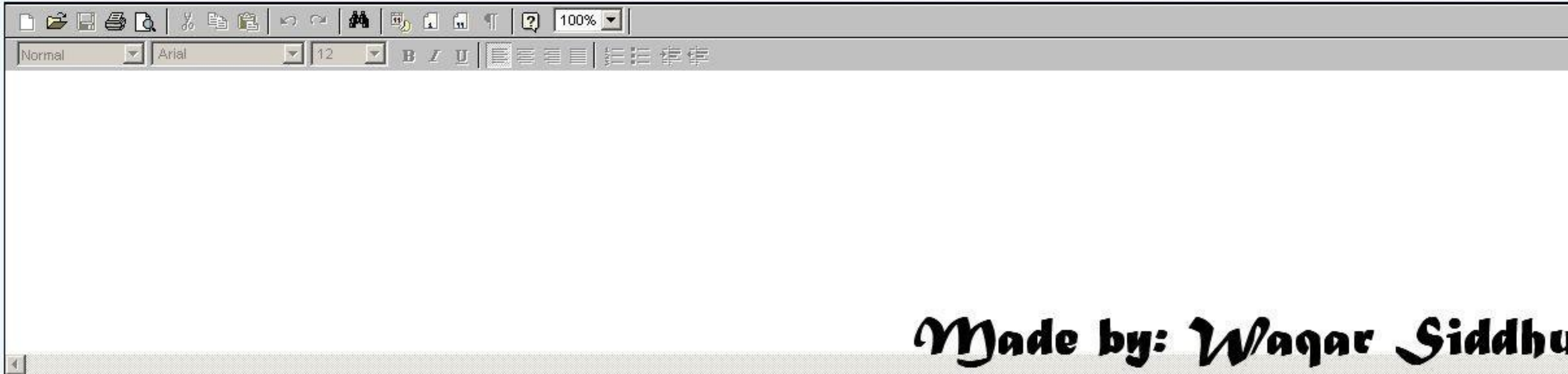

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

What is heap and heap order?

Answer (Please click here to Add Answer)

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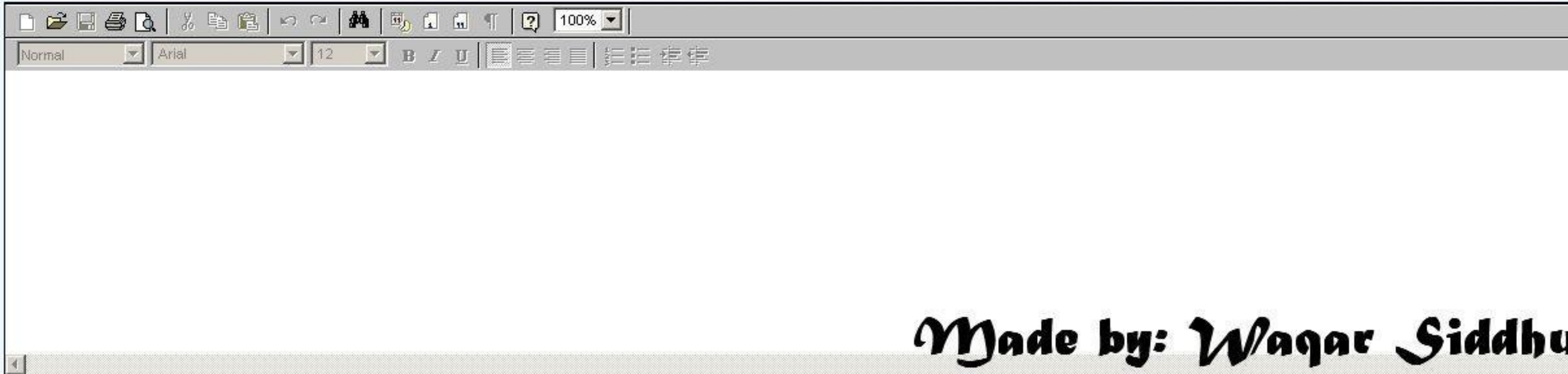

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Define free tree.

Answer (Please click here to Add Answer)

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A free tree is a tree with no
vertex designated as the root vertex. A free tree with n vertices has exactly n − 1 edges. There exists a unique path between any two vertices of a free tree. Adding any edge to a free tree creates a unique cycle. Breaking any edge on this cycle restores the free tree.

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The following adjacency matrix represents a graph that consists of four vertices labeled 0, 1, 2 and 3. The entries in the matrix indicate edge weights.

	0	1	2	3
0	0	1	0	3
1	2	0	4	0
2	0	1	0	1
3	2	0	0	0

Answer the following question:

Answer (Please click here to Add Answer)

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	0	1	2	3
0	0	1	0	3
1	2	0	4	0
2	0	1	0	1
3	2	0	0	0

Answer the following question:

Can an adjacency matrix for a directed graph ever *not* be square in shape? Why or why not?

Answer (Please click here to Add Answer)

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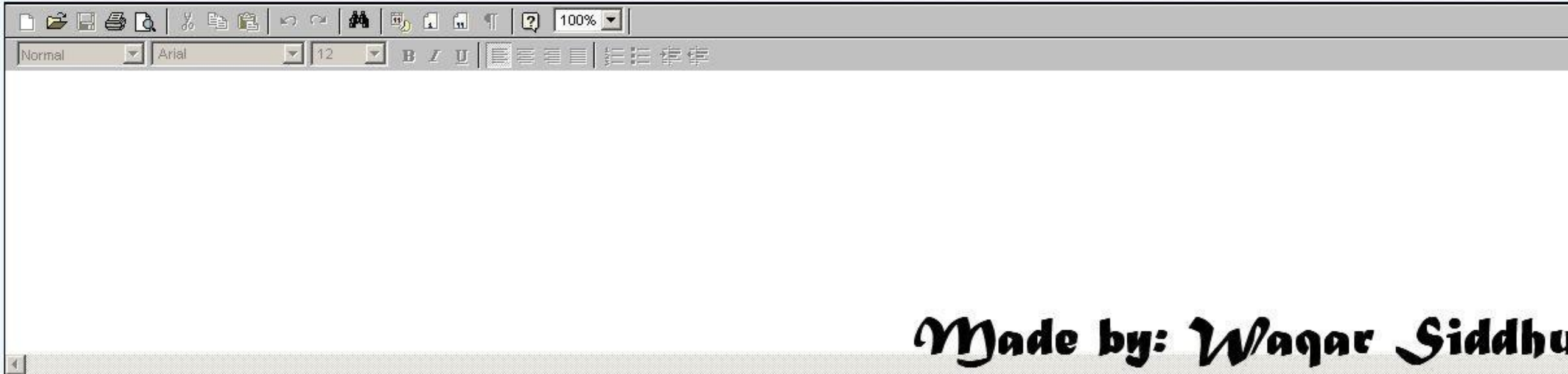

Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Where clique cover problem arises?

Answer (Please click here to Add Answer)

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The clique cover problem arises in applications of clustering. We put an edge between two nodes if they are similar enough to be clustered in the same group. We want to know whether it is possible to cluster all the vertices into k groups.

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Consider a digraph G = (V, E) and any DFS forest for G. Prove that G has a cycle if and only if the DFS forest has a back edge.

Answer (Please click here to Add Answer)

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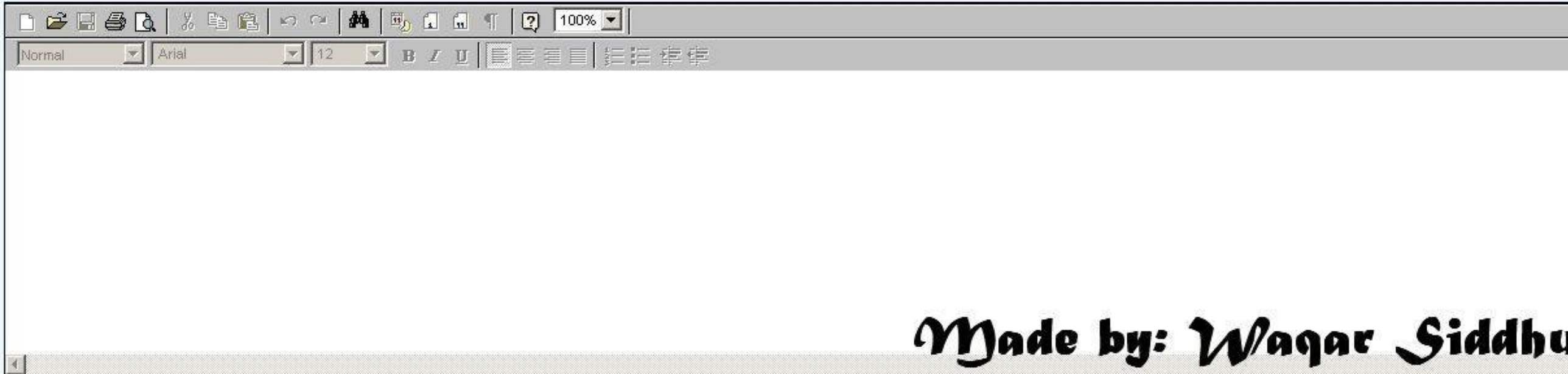

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Describe three asymptotic notations.

Answer (Please click here to Add Answer)

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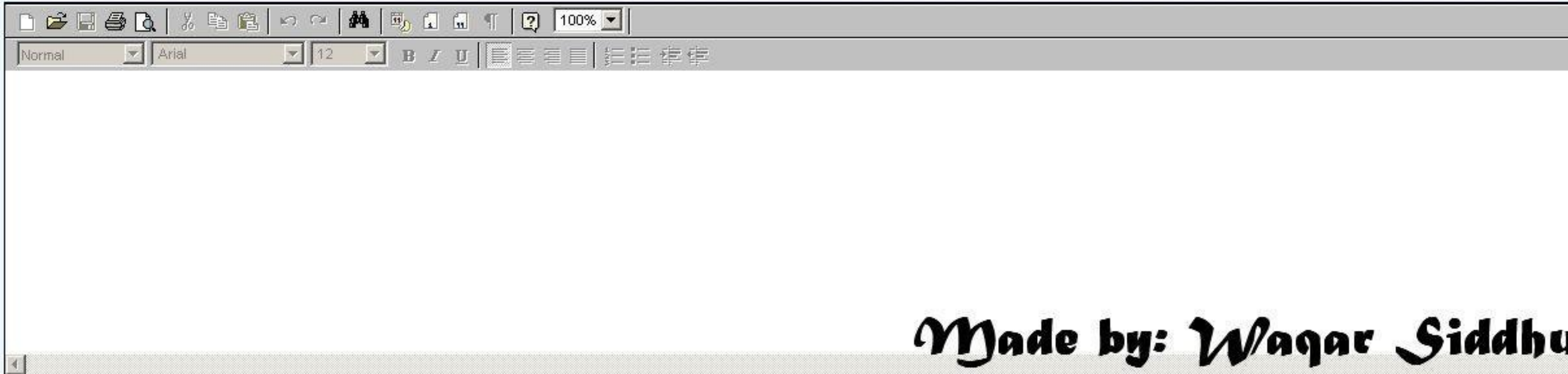

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Briefly discuss at least three variants of the shortest path problem.

Answer (Please click here to Add Answer)

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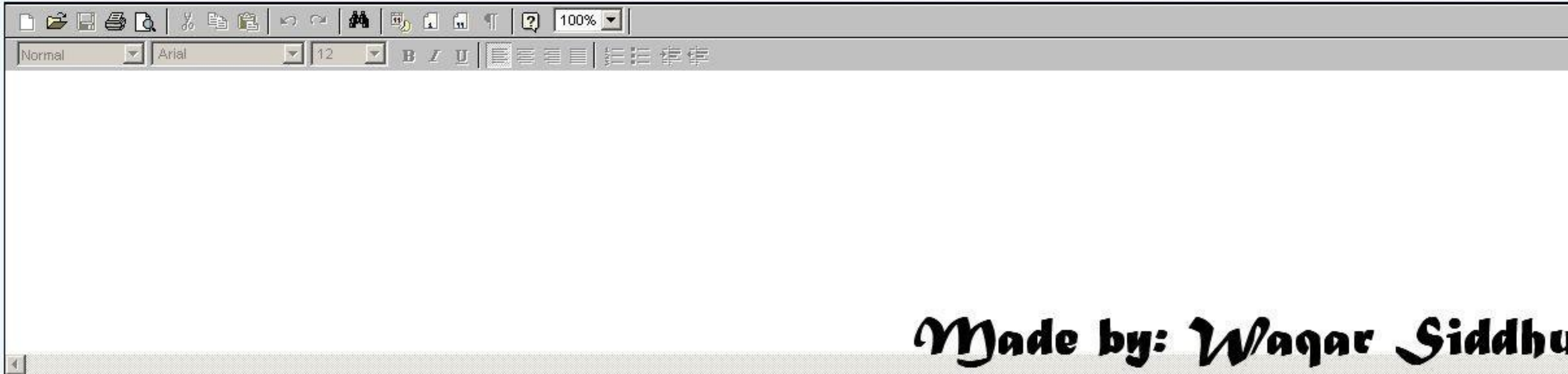

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What do you mean by polynomial time algorithm? Explain what kind of problem can be solved by using polynomial time algorithm?

Answer (Please click here to Add Answer)

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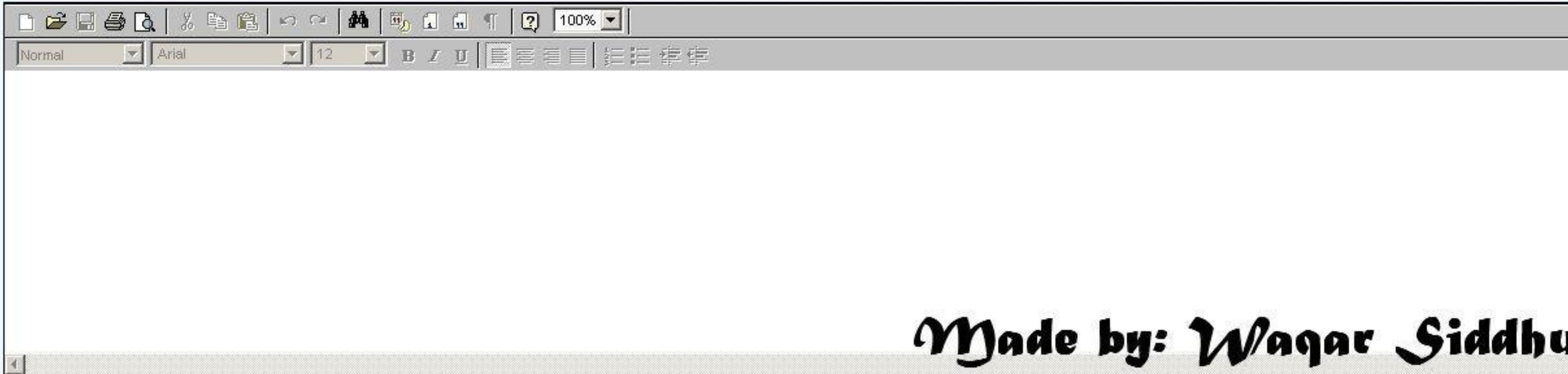

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Suppose you could reduce an NP-complete problem to a polynomial time problem in polynomial time. What would be the consequence?

Answer (Please click here to Add Answer)

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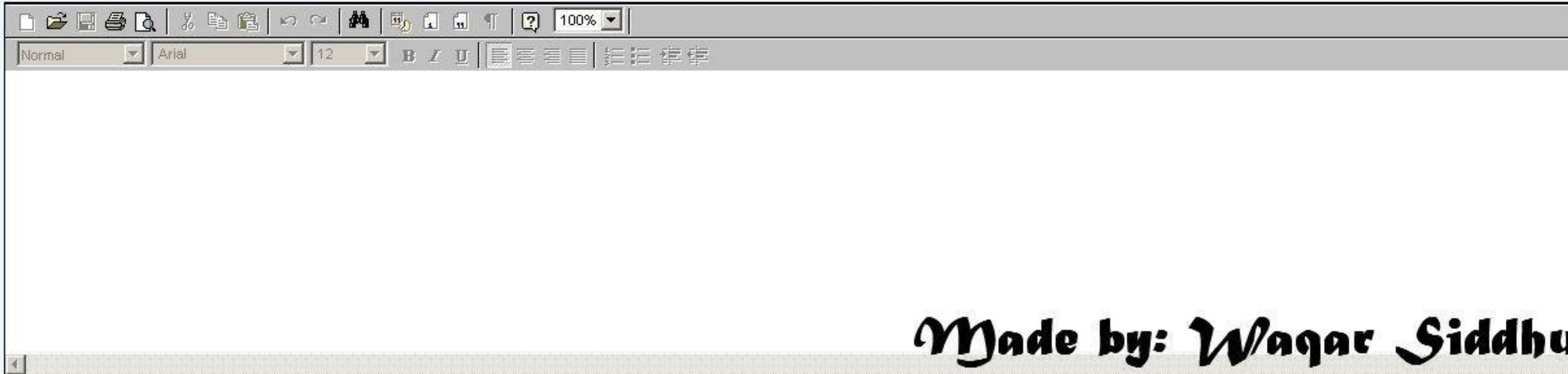
According to Dijkstra's Algorithm, write the pseudo code to relax a vertex.

Answer (Please click here to Add Answer)

VuAnswers.com



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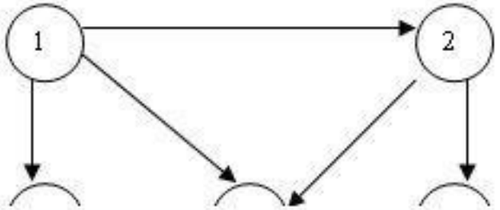
Find the adjacent list for the following graph



Answer (Please click here to Add Answer)

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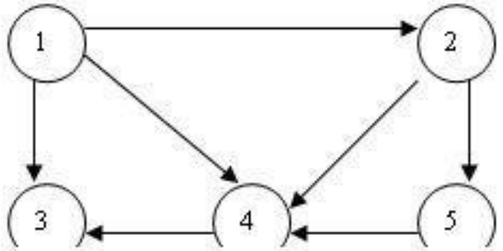

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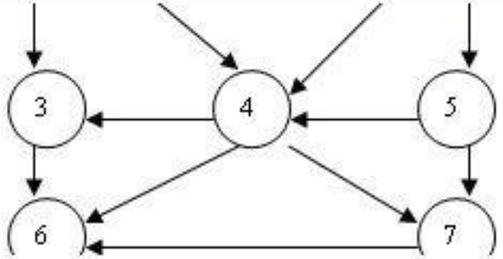

Answer (Please click here to Add Answer)

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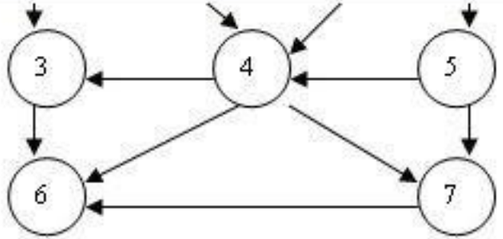

Answer (Please click here to Add Answer)

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Normal Arial 12 100%

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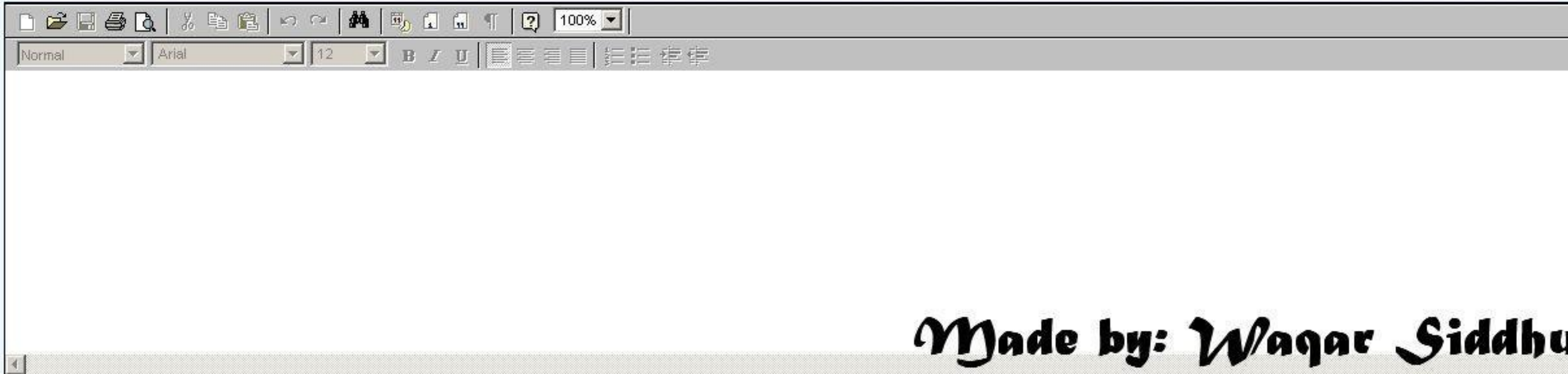
Draw the final **Max-Heap** structure for the following array,

`50, 31, 45, 30, 2, 7, 40, 12, 28, 1`

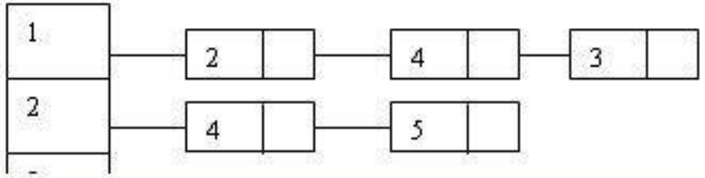
You can show the final result (tree) only.

Answer (Please click here to Add Answer)

VuAnswers.com



Consider the following adjacency list:



Answer (Please select your correct option)

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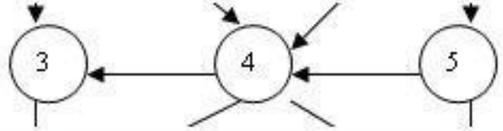
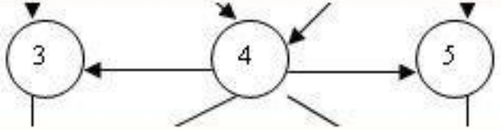
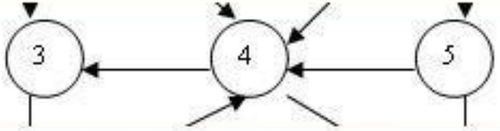
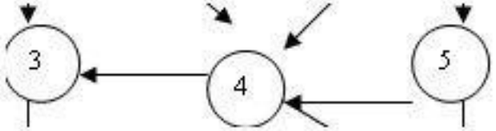
- 

- 

- 

- 

Made by: Waqar Siddhu

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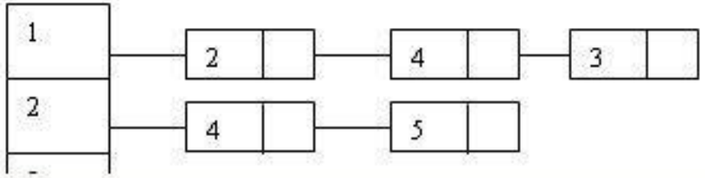

Answer (Please select your correct option)

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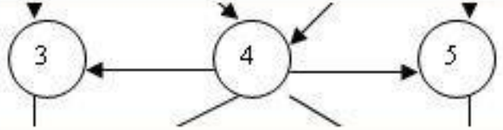

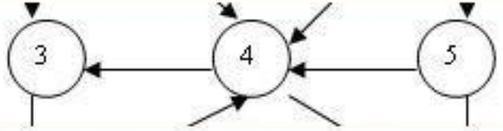

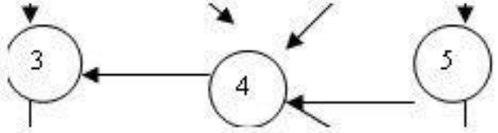

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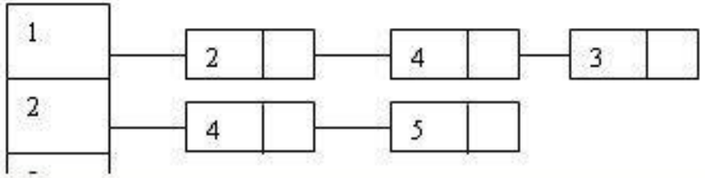

Answer (Please select your correct option)

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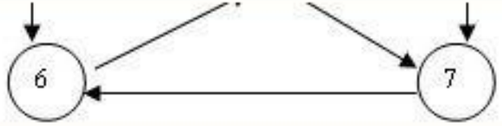

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

In strong components problem what complete refers to?

Answer (Please click here to Add Answer)

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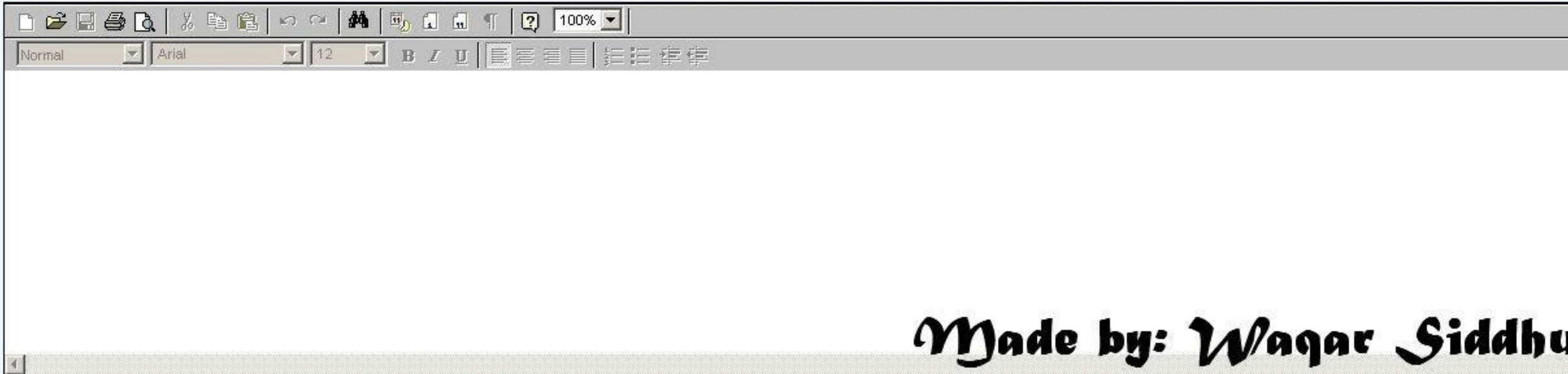
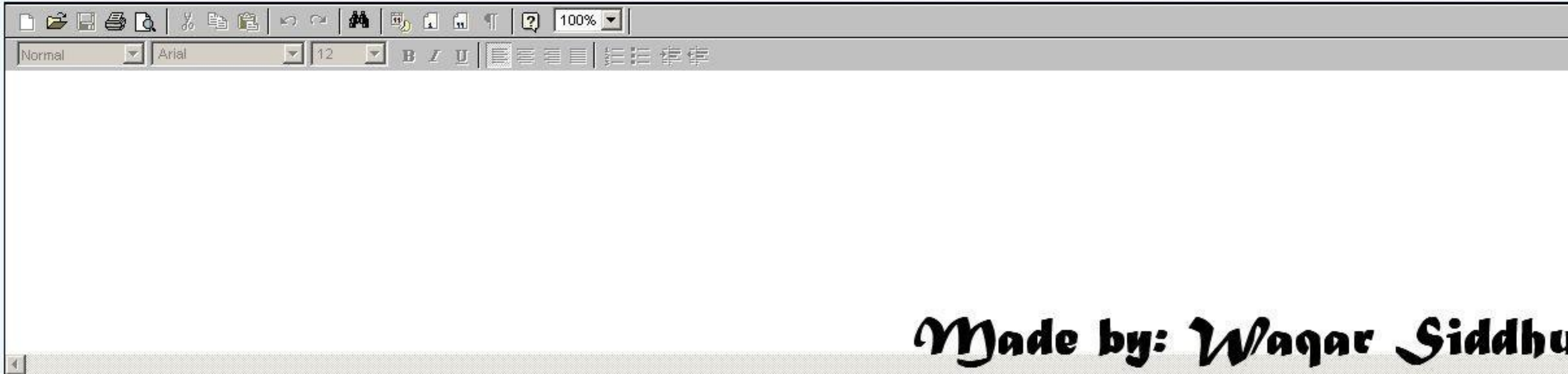


Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What are quadratic series?

Answer (Please click here to Add Answer)

VuAnswers.com



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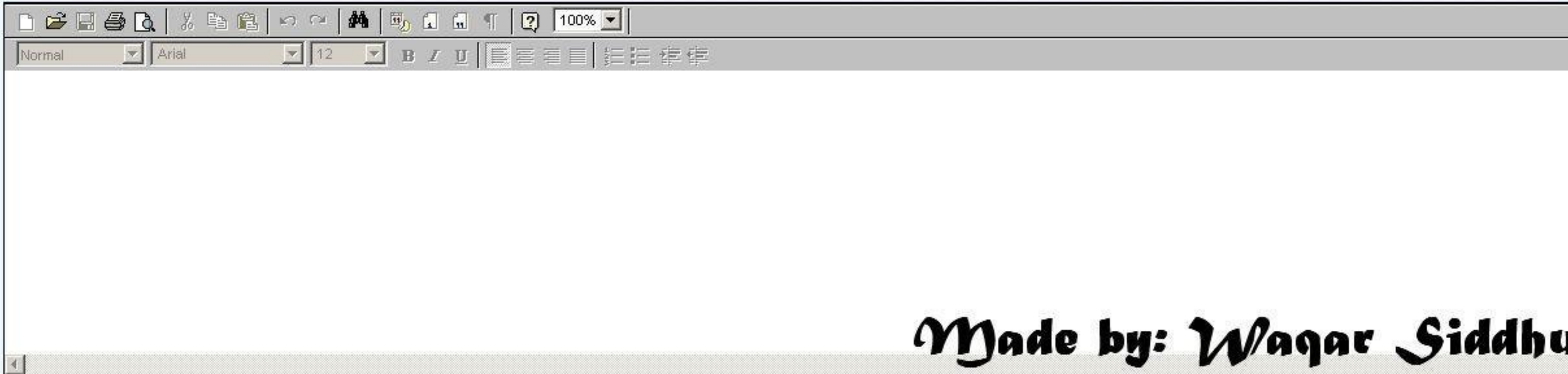
Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is overall time for Kruskal's algorithm if the graph is sparse?

Answer (Please click here to Add Answer)

VuAnswers.com



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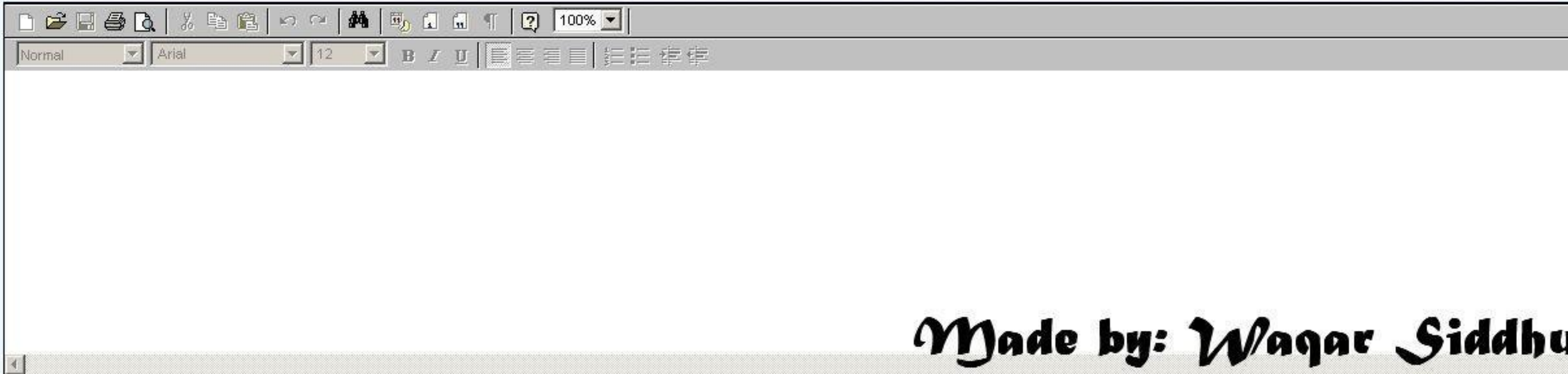
Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

When a problem is called a decision problem?

Answer (Please click here to Add Answer)

VuAnswers.com

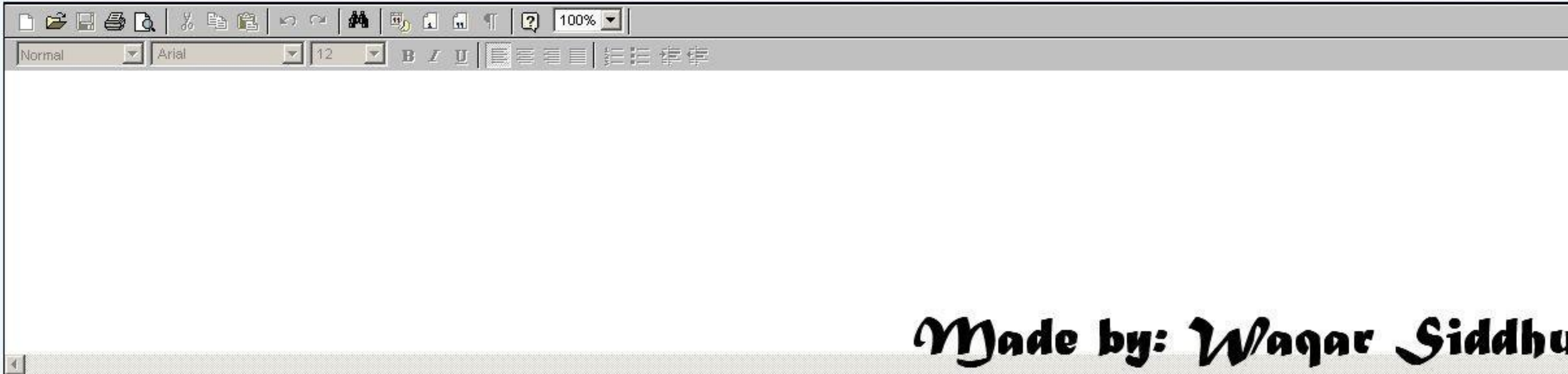


Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Formally describe Minimum Spanning Trees Problem.

Answer (Please click here to Add Answer)

VuAnswers.com



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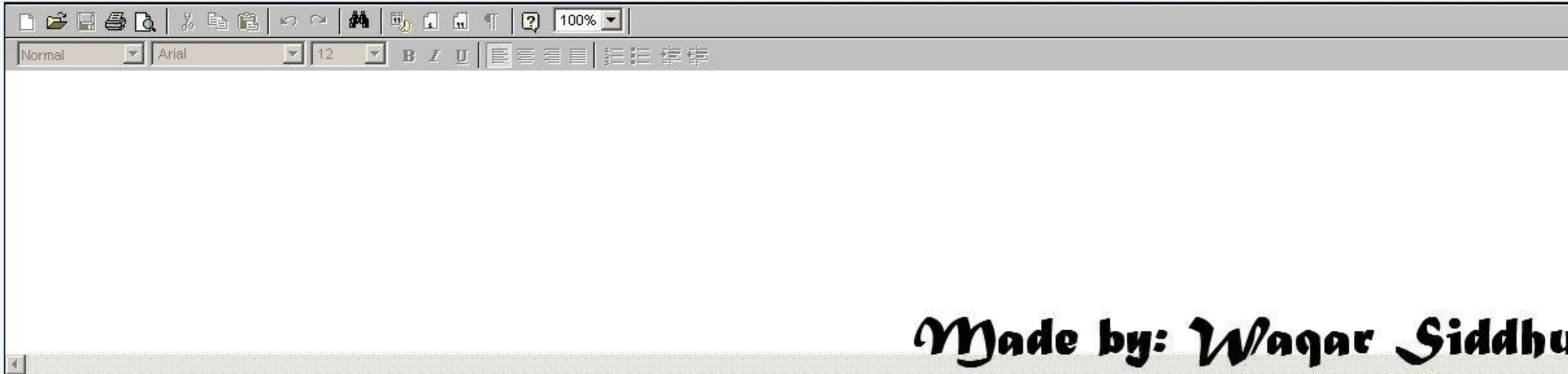
Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

True or false: A sequence of values in a column of the dynamic programming table for an instance of the knapsack problem is always non-decreasing? Give a brief description.

Answer (Please click here to Add Answer)

VuAnswers.com



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Describe the relation between mutually reachable, equivalence relation, and component digraph.

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

You are given the task of laying down new railway line between Peshawar and Karachi. There are *n* intermediate cities that can be used and you know the cost of laying track between any pair of thes cities. Your goal is to spend the least total amount of track to construct the railway line. How would you determine the least amount of track and the cities to go through? Name the best algorithm whic addresses the above problem.

Answer (Please click here to Add Answer)

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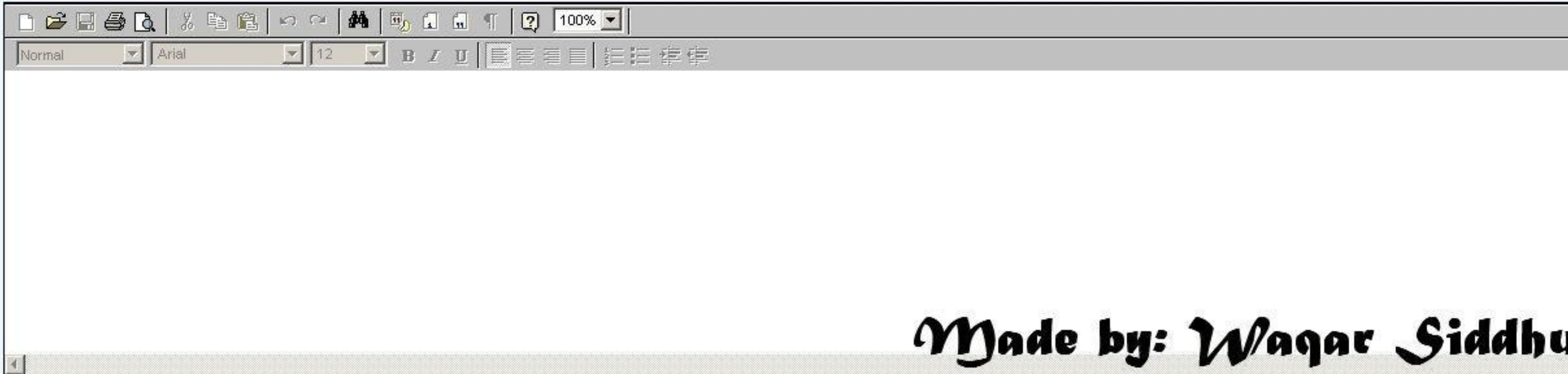

You are given the task of laying down new railway lines which will connect all *n* cities. Thus for any pair of cities, you will end up with track connecting them. Note that two routes may share the same track; track laid between Lahore and Islamabad can be used to travel in both directions. Your goal is to use the minimum amount of track. How would you achieve the goal now? (Note : consider the scenario carefully and name only the best suited algorithm)

1 Dijkstra's algorithm
2 Prims Algorithm
3 Folloyed Warshal Agorithm

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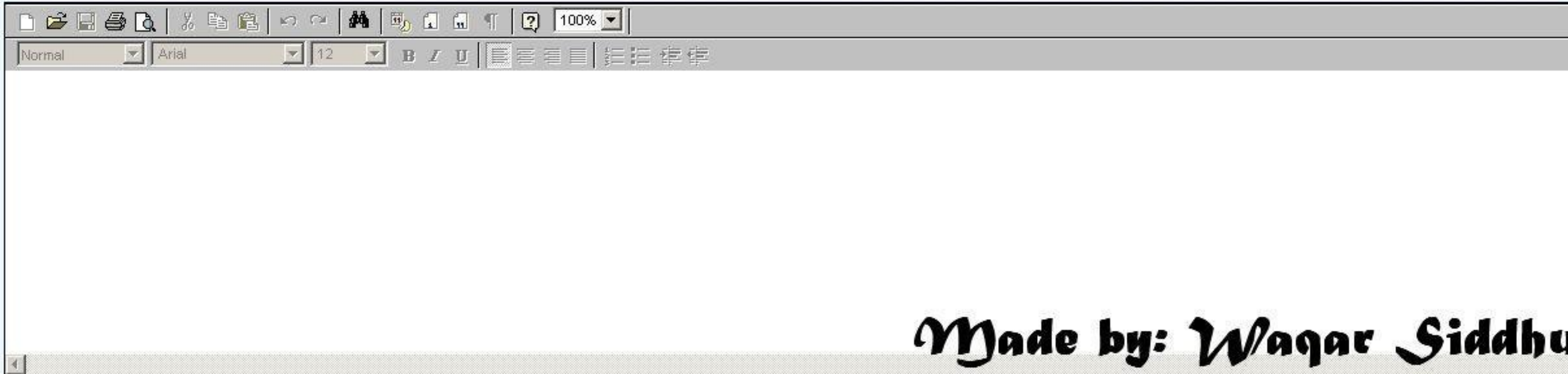
Answer (Please click here to Add Answer)



Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Following is not the minimum spanning tree convert it into MST. [You can show final result in exam software and need not to show all intermediate steps].

Answer (Please click here to Add Answer)

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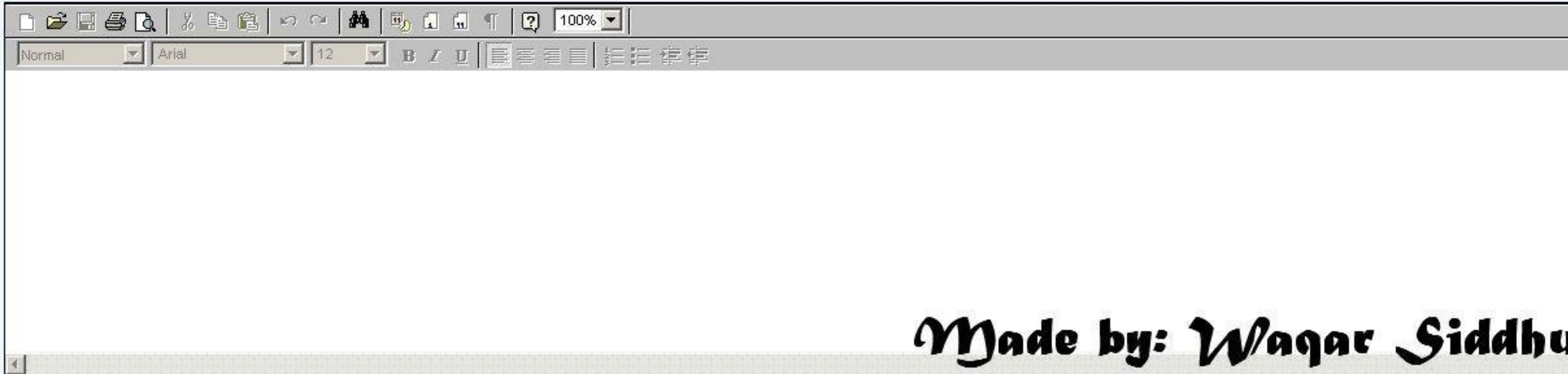
track; track laid between Lahore and Islamabad can be used to travel in both directions. Your goal is to use the minimum amount of track. How would you achieve the goal now? (Note : consider the scenario carefully and name only the best suited algorithm)

1 Dijkstra's algorithm
2 Prims Algorithm
3 Folloyed Warshal Agorithm
4 Bellman Ford Algorithm.

Answer (Please click here to Add Answer)

VuAnswers.com



Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Following is not the minimum spanning tree convert it into MST. [You can show final result in exam software and need not to show all intermediate steps].

Answer (Please click here to Add Answer)

VuAnswers.com

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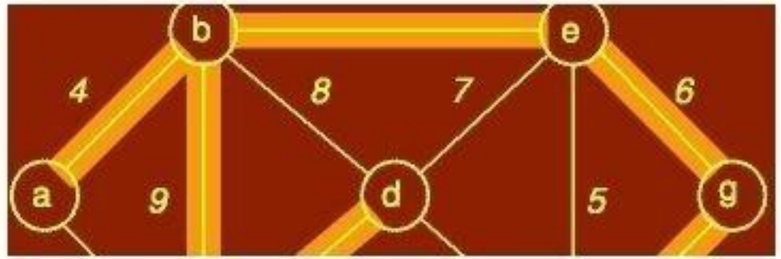

Answer (Please click here to Add Answer)

VuAnswers.com

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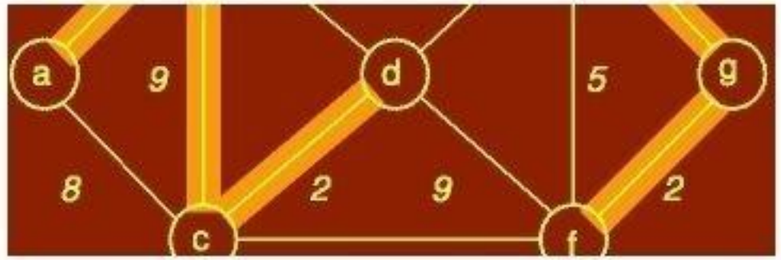

Answer (Please click here to Add Answer)

VuAnswers.com

Made by: Waqar Siddhu

Answer (Please click here to Add Answer)

VuAnswers.com

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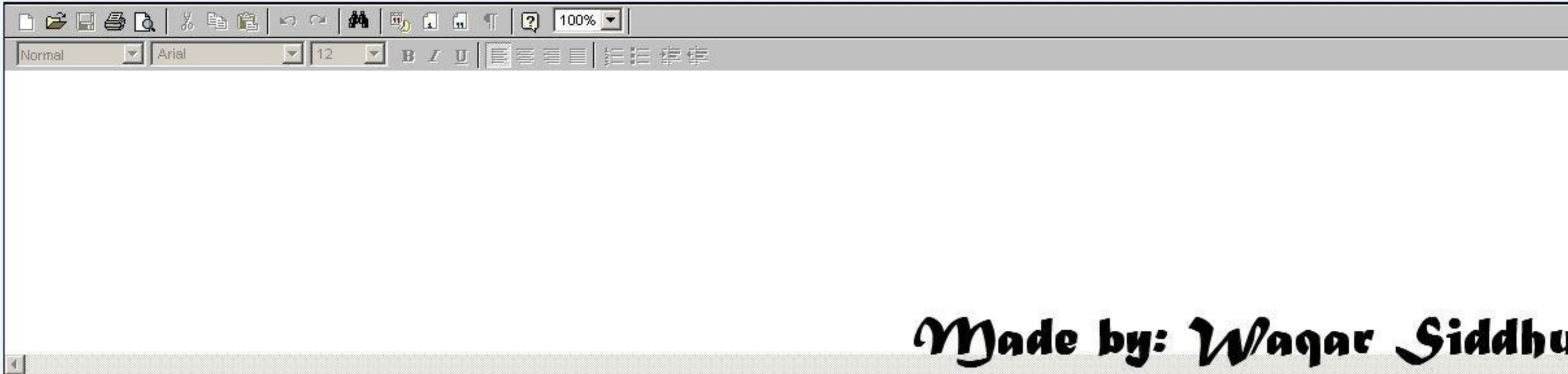
Consider the following recursive `search` function which returns the index of the `array` element containing `key`, if such an element exists. Find out what is the complexity of search and show you work.

```
int search( int* array, int left, int right, int key){
     int mid = (left + right)/2;
if( left == right )
     return left;
```

Answer (Please click here to Add Answer)

VuAnswers.com



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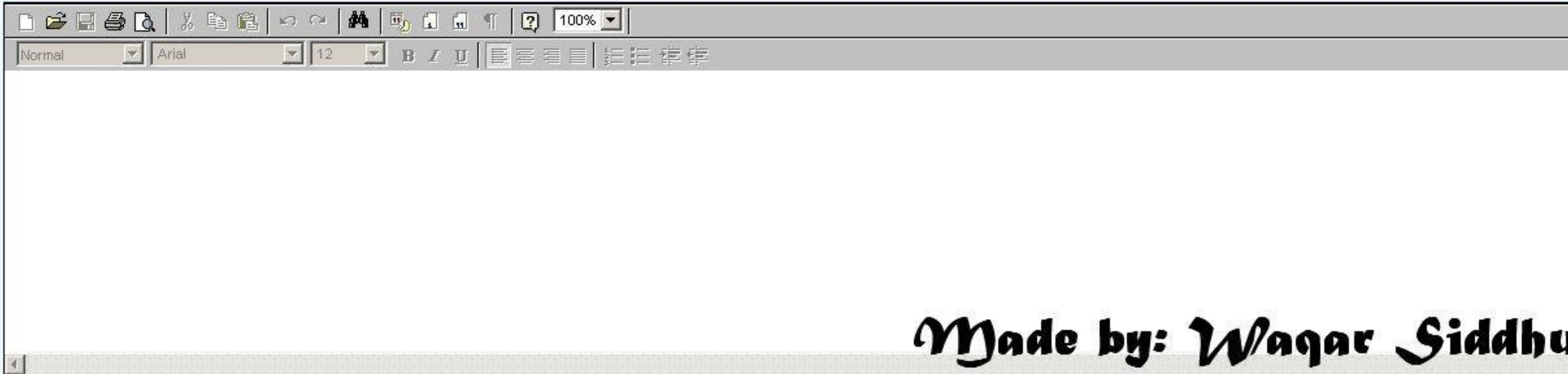
```
    return left;
else if( array[mid] <= key )
    return search( array, mid+1, right, key );
else
    return search( array, left, mid, key );
}
```

Answer (Please click here to Add Answer)

VuAnswers.com



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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Write pseudo code for the algorithm, if we implement the bag of knapsack by using a stack.

Answer (Please click here to Add Answer)

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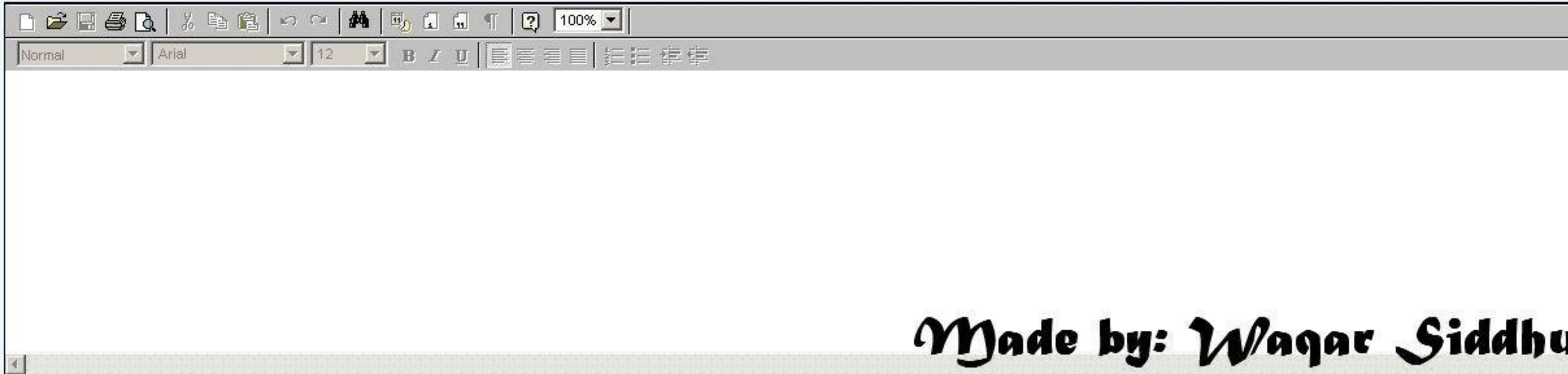

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